Register Transfer and Microoperations

Contents

1-1 Register Transfer Language (RTL)

- **Digital System**: An interconnection of hardware modules that do a certain task on the information.
- Registers + Operations performed on the data stored in them = **Digital Module**
- Modules are interconnected with common data and control paths to form a digital computer system

1-1 Register Transfer Language
cont.

- **Microoperations**: operations executed on data stored in one or more registers.
- For any function of the computer, a sequence of microoperations is used to describe it
- The result of the operation may be:
  - replace the previous binary information of a register or
  - transferred to another register

101101110011 $\xrightarrow{\text{Shift Right Operation}}$ 010110111001

1-1 Register Transfer Language cont.

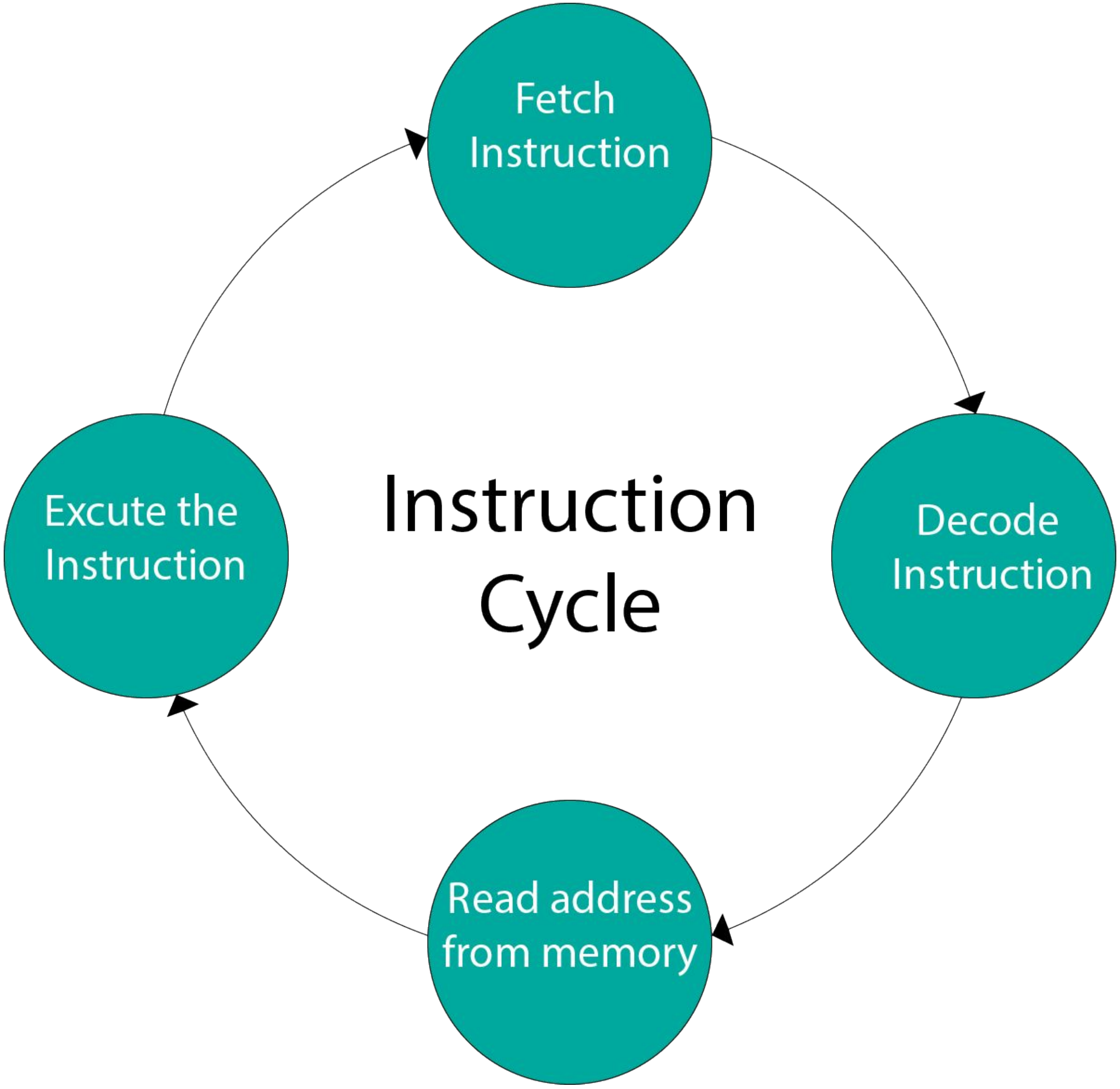
- The internal hardware organization of a digital computer is defined by specifying:
  - The **set of registers** it contains and their function
  - The **sequence of microoperations** performed on the binary information stored in the registers
  - The **control** that initiates the sequence of microoperations
- Registers + Microoperations Hardware + Control Functions = **Digital Computer**

1-1 Register Transfer Language
cont.

- Register Transfer Language (RTL) : a symbolic notation to describe the microoperation transfers among registers

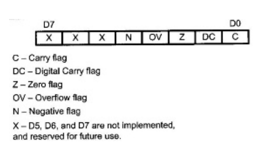
Next steps:

   - Define symbols for various types of microoperations,
   - Describe the hardware that implements these microoperations


Fetch Instruction
Decode Instruction
Read address from memory
Excute the Instruction
Instruction Cycle

1-2 Register Transfer (our first microoperation)

- Computer registers are designated by capital letters (sometimes followed by numerals) to denote the function of the register
  - R1: processor register
  - MAR: Memory Address Register (holds an address for a memory unit)
  - PC: Program Counter (each instruction gets fetched, the **program counter** increases its stored value by 1)
  - IR: Instruction Register (that holds the **instruction** currently being executed or decoded)
  - SR: Status Register



1-2 Register Transfer cont.

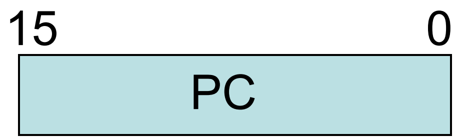
- The individual flip-flops in an n-bit register are numbered in sequence from 0 to n-1 **(from the right position toward the left position)**

Register R1 Showing individual bits

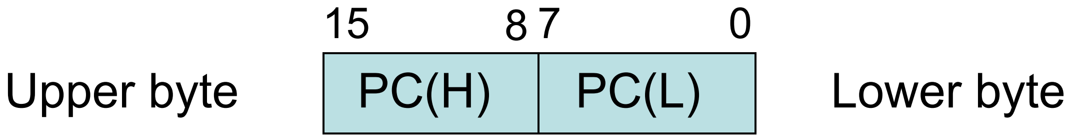
A block diagram of a register

1-2 Register Transfer $^{cont.}$

Other ways of drawing the block diagram of a register:



Numbering of bits



Partitioned into two parts

1-2 Register Transfer cont.

- Information transfer from one register to another is described by a *replacement operator:* $\mathbf{R2} \leftarrow \mathbf{R1}$
- This statement denotes a transfer of the content of register R1 into register R2
- **The transfer happens in one clock cycle**
- The content of the R1 (source) does not change
- The content of the R2 (destination) will be lost and replaced by the new data transferred from R1
- We are assuming that the circuits are available from the outputs of the source register to the inputs of the destination register, and that the destination register has a parallel load capability

1-2 Register Transfer cont.

- Conditional transfer occurs only under a control condition

- Representation of a (conditional) transfer

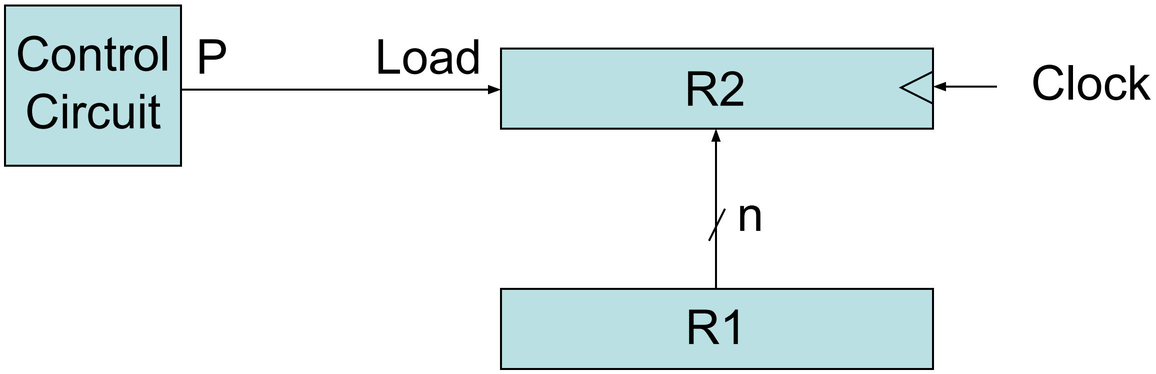
  $P: \quad R2 \leftarrow R1$

- A binary condition (P equals to 0 or 1) determines when the transfer occurs
- The content of R1 is transferred into R2 only if P is 1

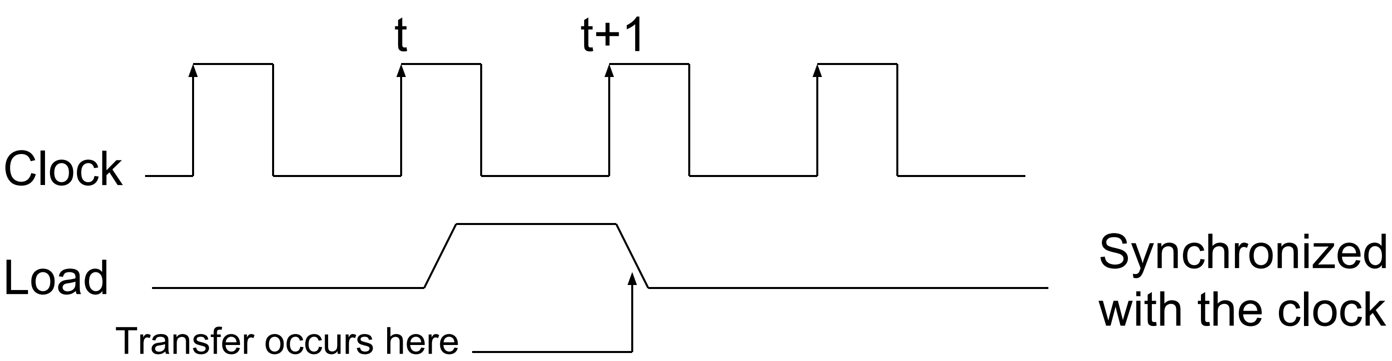
1-2 Register Transfer cont.

Hardware implementation of a controlled transfer: P: R2 ← R1

Block diagram:



Timing diagram



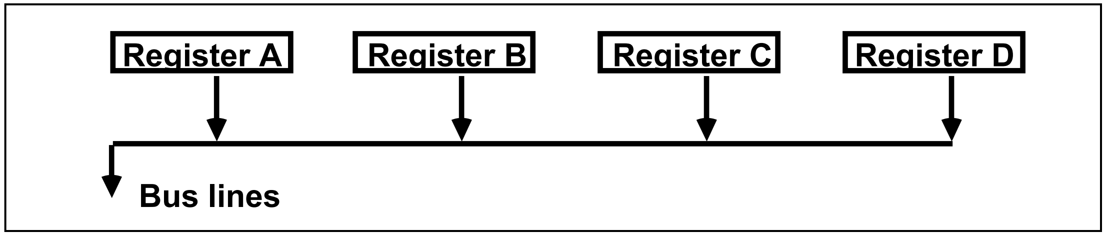
1-2 Register Transfer cont.

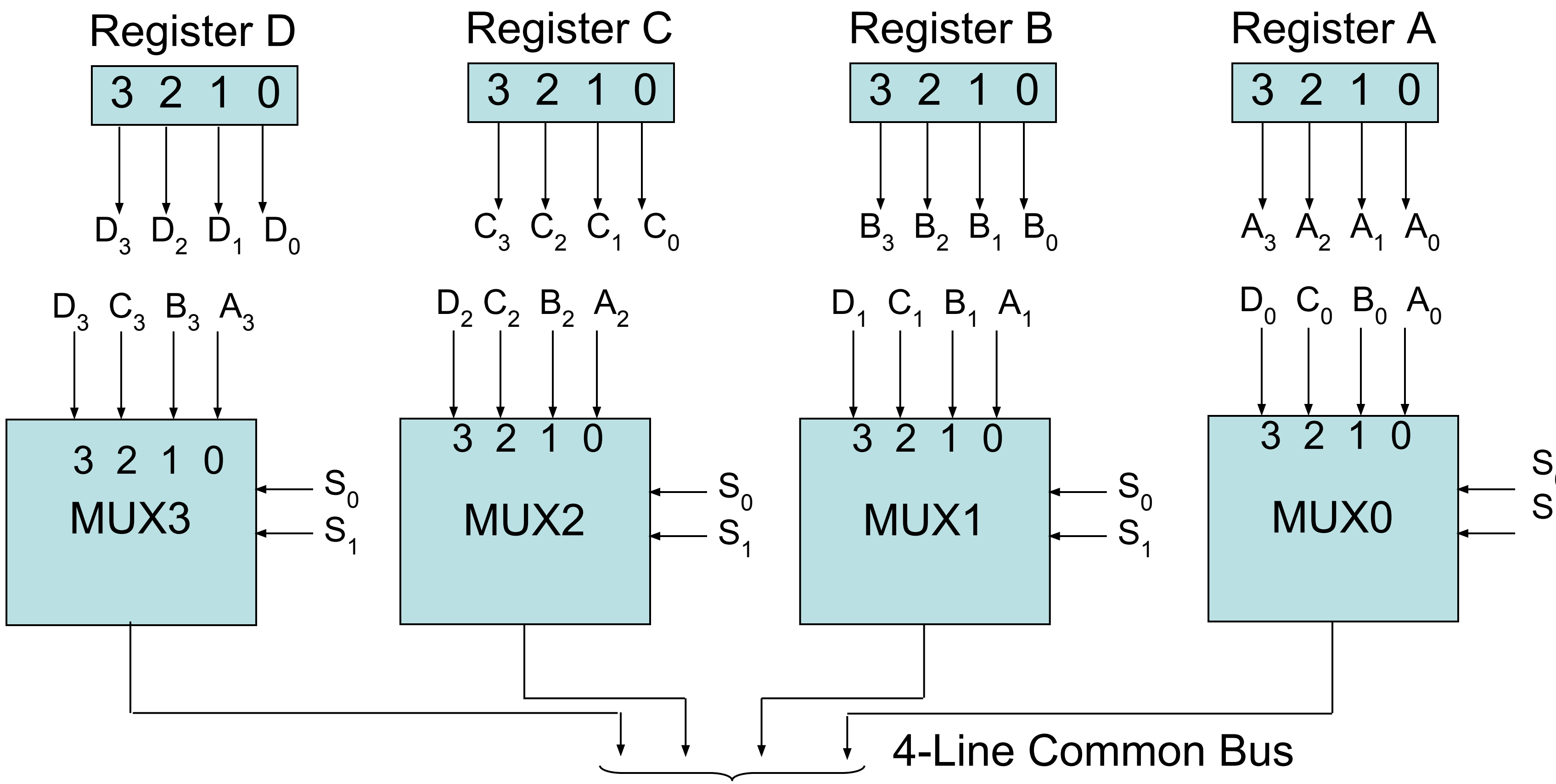
Basic Symbols for Register Transfers		
Symbol	Description	Examples
Letters & numerals	Denotes a register	MAR, R2
Parenthesis ( )	Denotes a part of a register	R2(0-7), R2(L)
Arrow $\leftarrow$	Denotes transfer of information	R2 $\leftarrow$ R1
Comma ,	Separates two microoperations	R2 $\leftarrow$ R1, R1 $\leftarrow$ R2

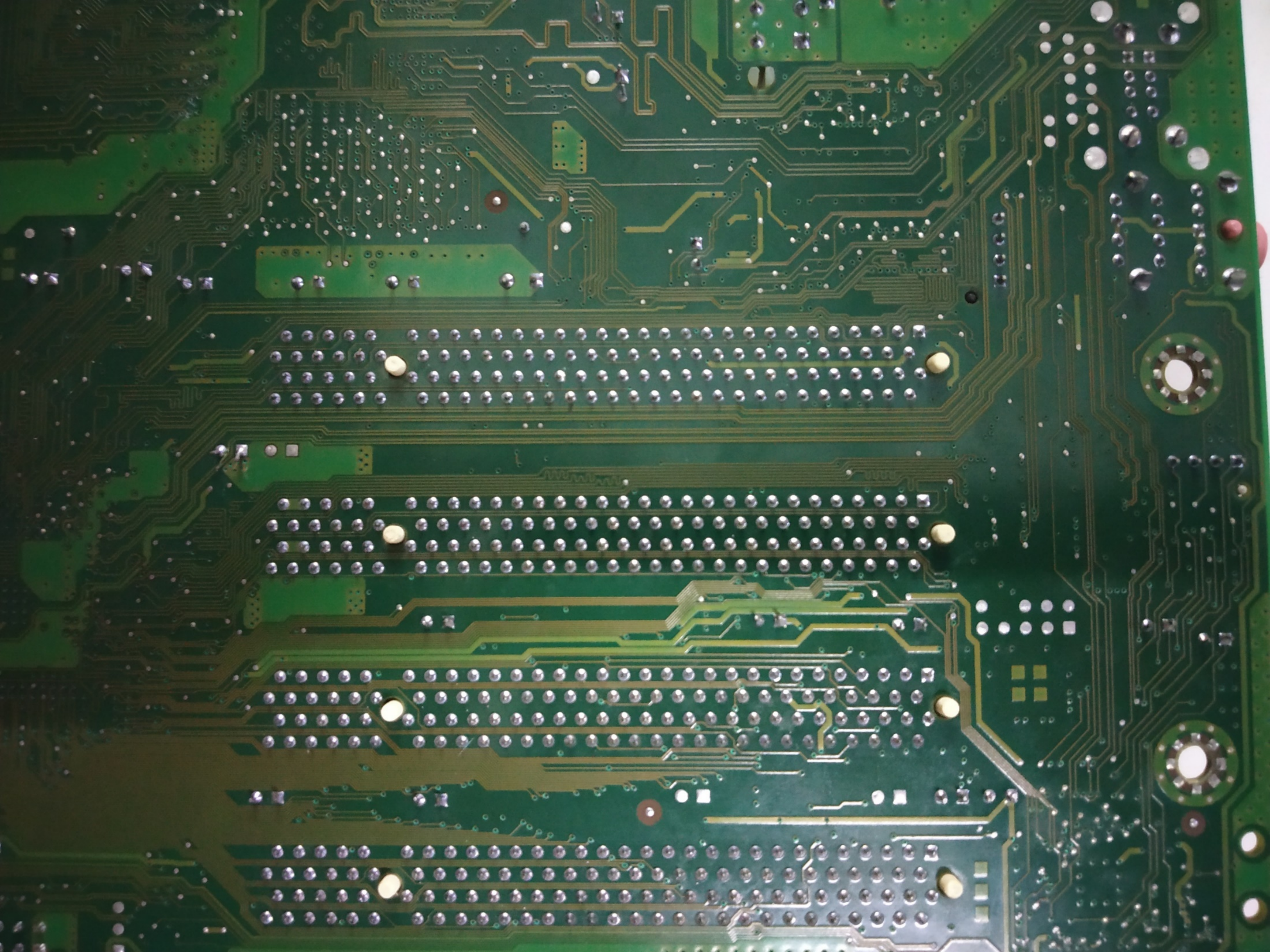
1-3 Bus and Memory Transfers

- **Paths** must be provided to transfer information from one register to another
- A ***Common Bus System*** is a scheme for transferring information between registers in a multiple-register configuration
- **A bus**: set of common lines, one for each bit of a register, through which binary information is transferred one at a time
- **Control signals** determine which **register is selected** by the bus during each particular register transfer

1-3 Bus and Memory Transfers







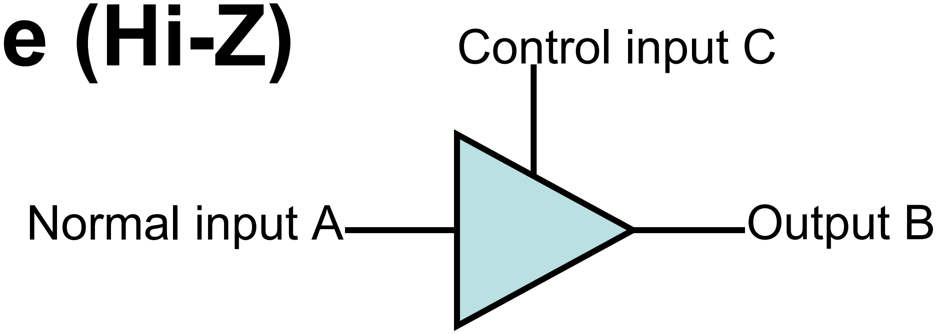
1-3 Bus and Memory Transfers

- The transfer of information from a bus into one of many destination registers is done:
  - By **connecting the bus lines to the inputs of all destination registers and then**:
  - **activating the load control of the particular destination register selected**
- We write: $R2 \leftarrow C$ to symbolize that the content of register C is *loaded into* the register R2 using the common system bus
- It is equivalent to: $BUS \leftarrow C$, (select C)

  $R2 \leftarrow BUS$ (Load R2)

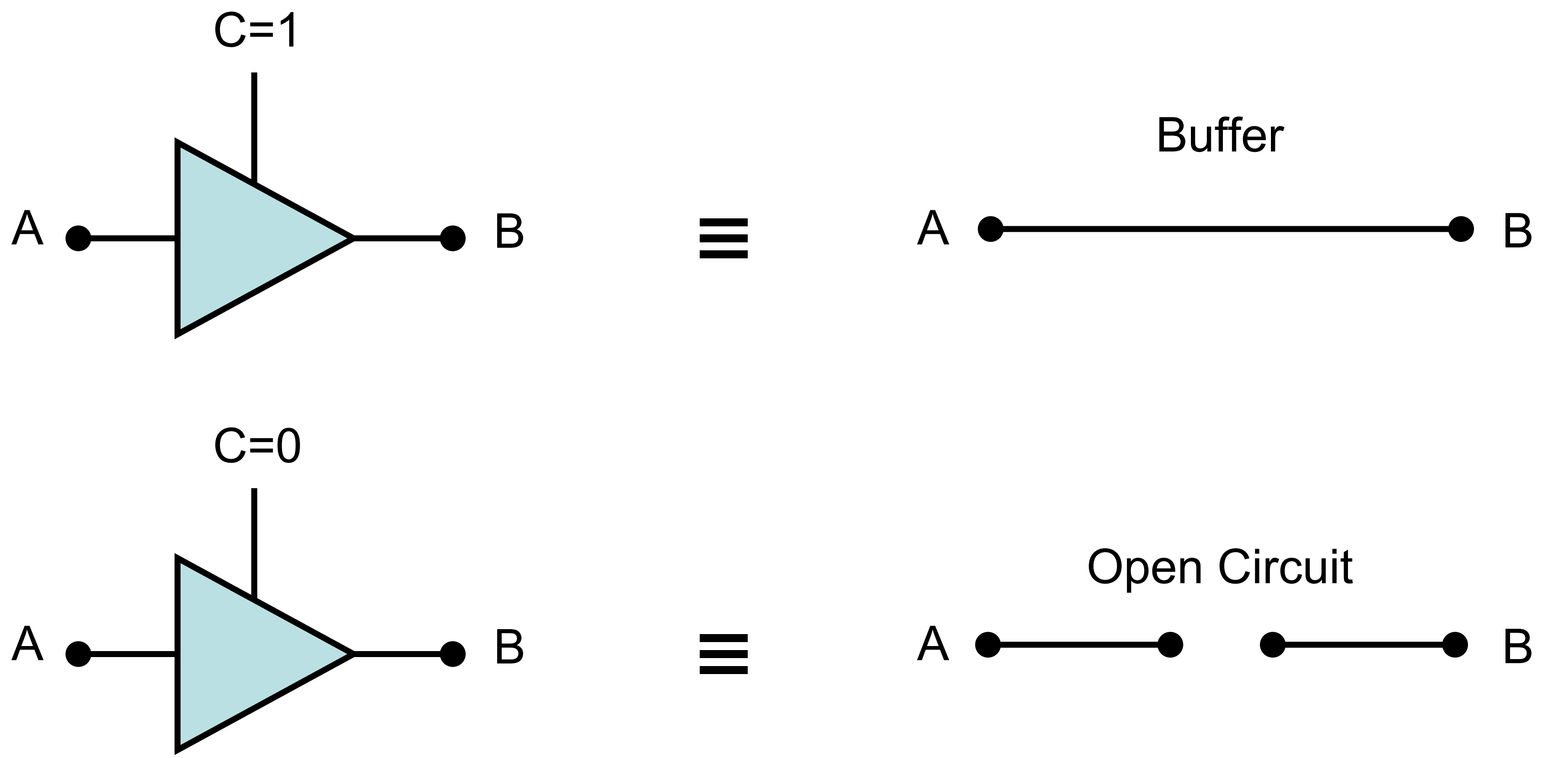
1-3 Bus and Memory Transfers: **Three-State Bus Buffers**

- A bus system can be constructed with three-state buffer gates instead of multiplexers
- A three-state buffer is a digital circuit that exhibits three states: **logic-0**, **logic-1**, and **high-impedance (Hi-Z)**

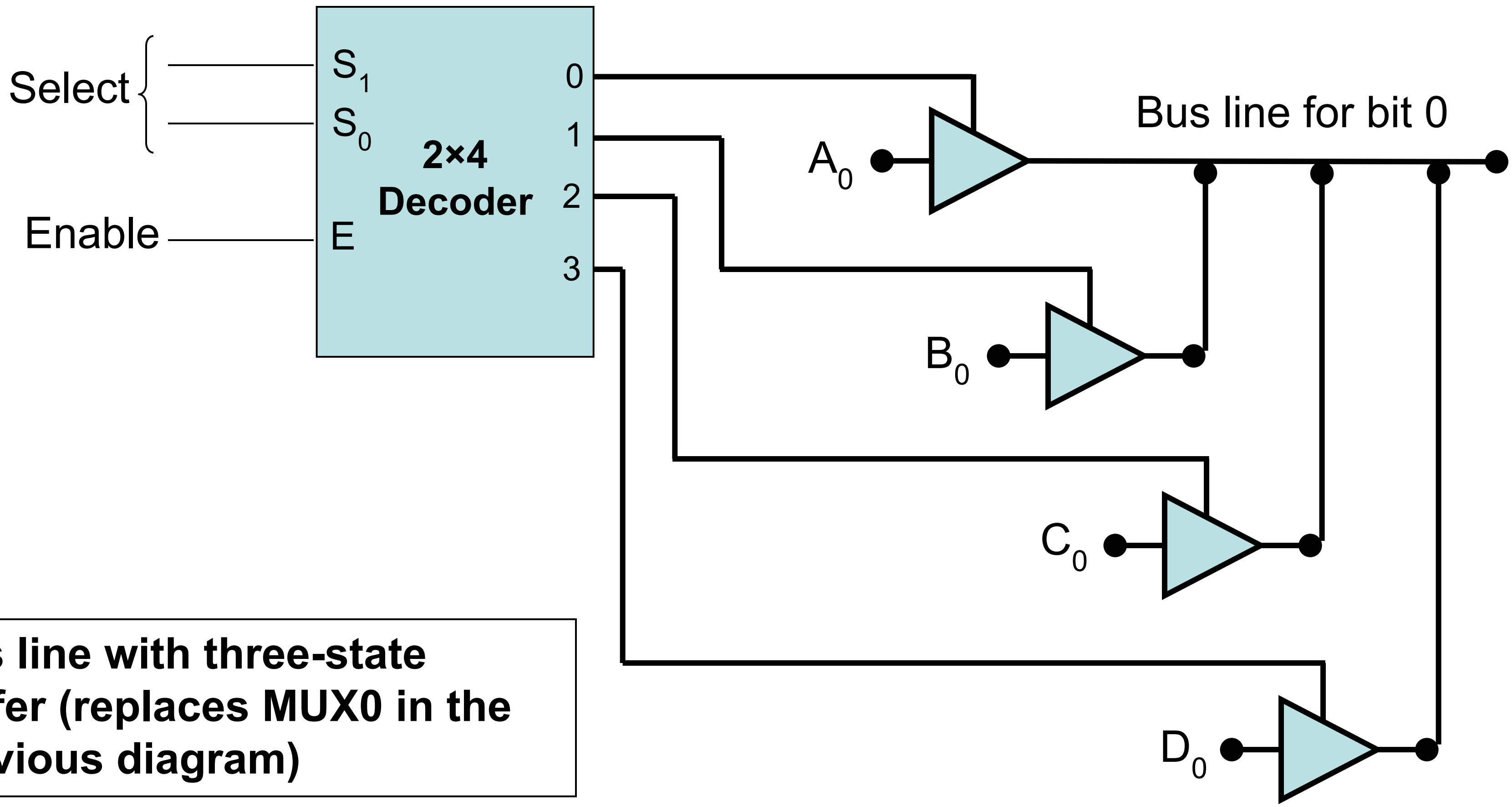


Three-State Buffer

1-3 Bus and Memory Transfers: Three-State Bus Buffers cont.



1-3 Bus and Memory Transfers: Three-State Bus Buffers cont.



Bus line with three-state buffer (replaces MUX0 in the previous diagram)

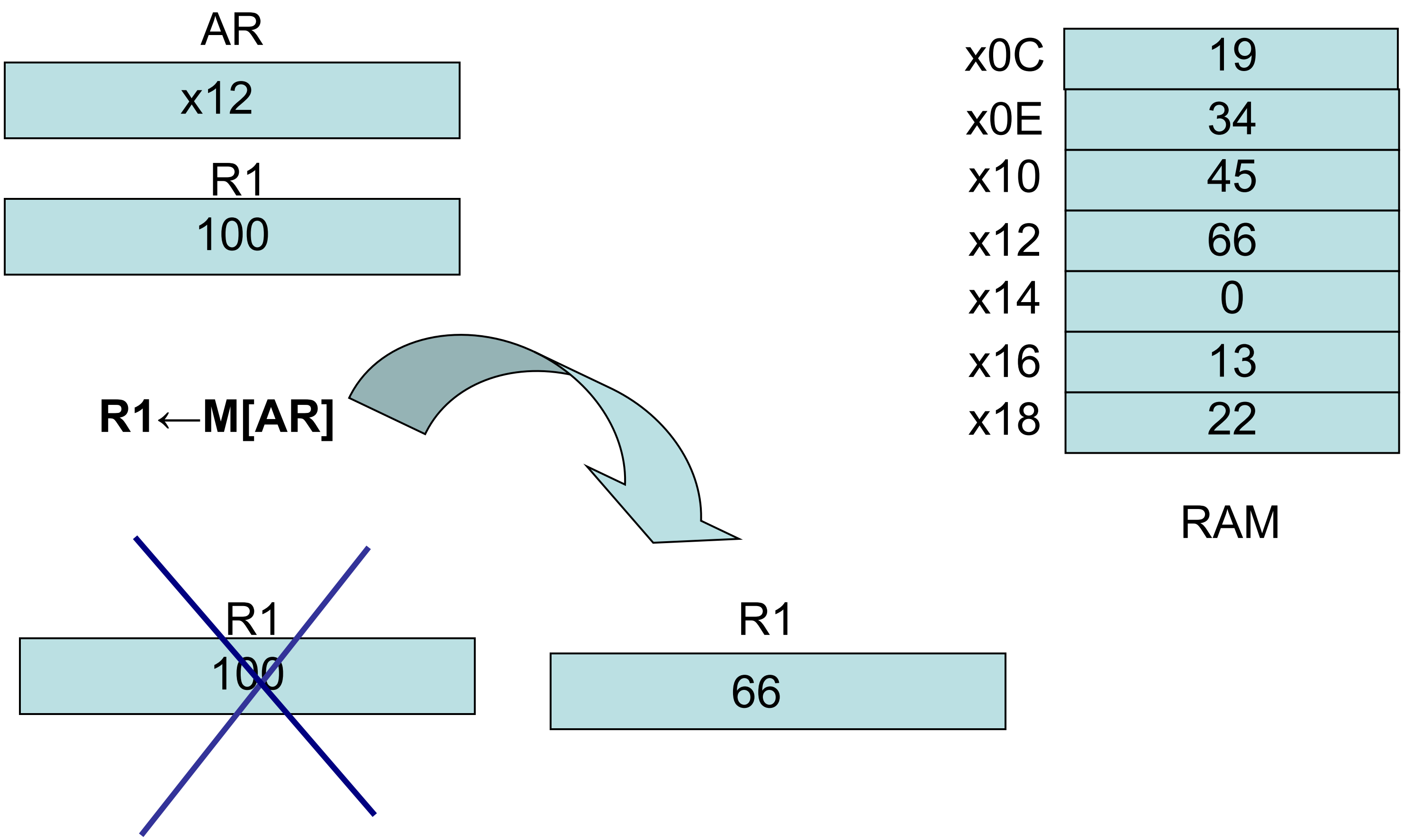
1-3 Bus and Memory Transfers: Memory Transfer

- **Memory read** : Transfer from memory
- **Memory write** : Transfer to memory
- Data being read or wrote is called a **memory word** (called M)- (refer to section 2-7)
- It is necessary to specify the address of M when writing /reading memory
- This is done by enclosing the address in square brackets following the letter M
- Example: M[0016] : the memory contents at address 0x0016

1-3 Bus and Memory Transfers: Memory Transfer cont.

- Assume that the address of a memory unit is stored in a register called the Address Register **AR**
- Lets represent a Data Register with DR, then:
  - **Read: DR $\leftarrow$ M[AR]**
  - **Write: M[AR] $\leftarrow$ DR**

1-3 Bus and Memory Transfers: Memory Transfer cont.



1-4 Arithmetic Microoperations

- The microoperations most often encountered in digital computers are classified into four categories:
  - **Register transfer microoperations**
  - **Arithmetic microoperations (on numeric data stored in the registers)**
  - **Logic microoperations (bit manipulations on non-numeric data)**
  - **Shift microoperations**

1-4 Arithmetic Microoperations cont.

- The basic arithmetic microoperations are: **addition, subtraction, increment, decrement**, and **shift**
- Addition Microoperation:

$$R3 \leftarrow R1+R2$$

- Subtraction Microoperation:

$$R3 \leftarrow R1-R2 \text{ or :}$$

1's complement

$$R3 \leftarrow R1+\overline{R2}+1$$

1-4 Arithmetic Microoperations cont.

- One's Complement Microoperation:

$$\mathbf{R2 \leftarrow \overline{R2}}$$

- Two's Complement Microoperation:

$$\mathbf{R2 \leftarrow \overline{R2}+1}$$

- Increment Microoperation:

$$\mathbf{R2 \leftarrow R2+1}$$

- Decrement Microoperation:

$$\mathbf{R2 \leftarrow R2-1}$$

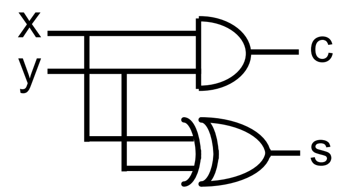
Half Adder/Full Adder

Half Adder

x	y	c	s
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

$c = xy$

$s = xy' + x'y$
$= x \oplus y$



Full Adder

x	y	c_{n-1}	c_n	s
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

c_n (K-map; y over the right column, x beside the lower two rows, c_{n-1} beside the middle two rows):

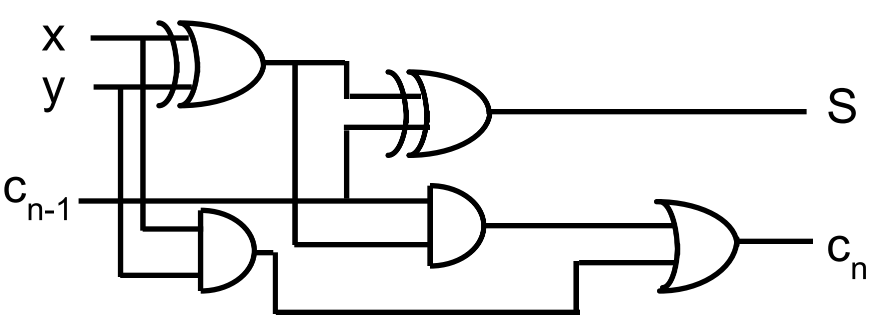
0	0	
0	1	c_{n-1}
1	1	c_{n-1}
0	1	

s (K-map; same layout):

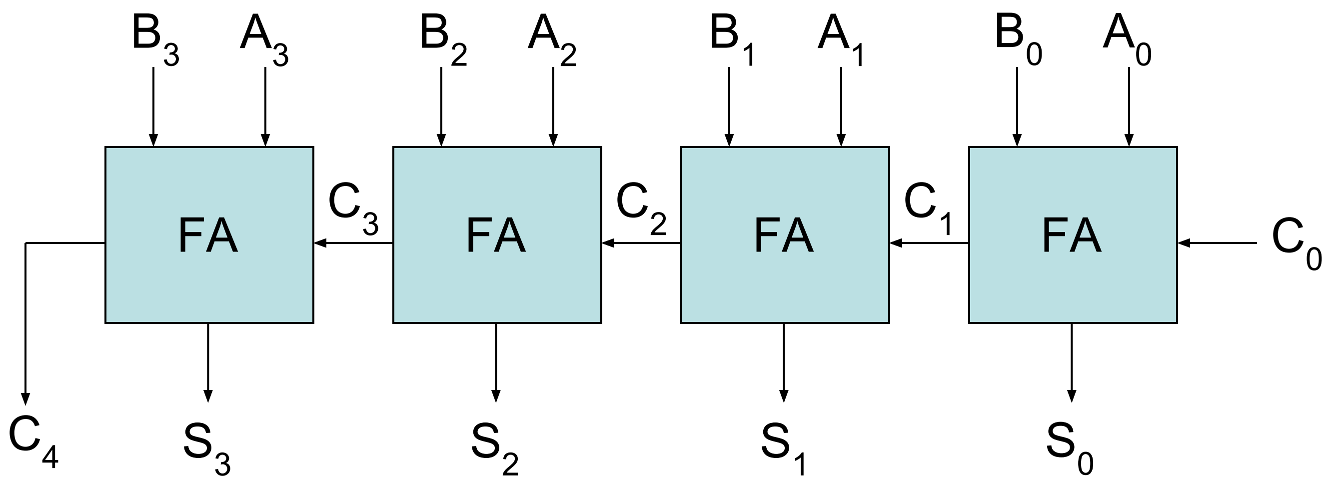
0	1	
1	0	c_{n-1}
0	1	c_{n-1}
1	0	

$c_n = xy + xc_{n-1} + yc_{n-1}$
$= xy + (x \oplus y)c_{n-1}$

$s = x'y'c_{n-1} + x'yc'_{n-1} + xy'c'_{n-1} + xyc_{n-1}$
$= x \oplus y \oplus c_{n-1} = (x \oplus y) \oplus c_{n-1}$

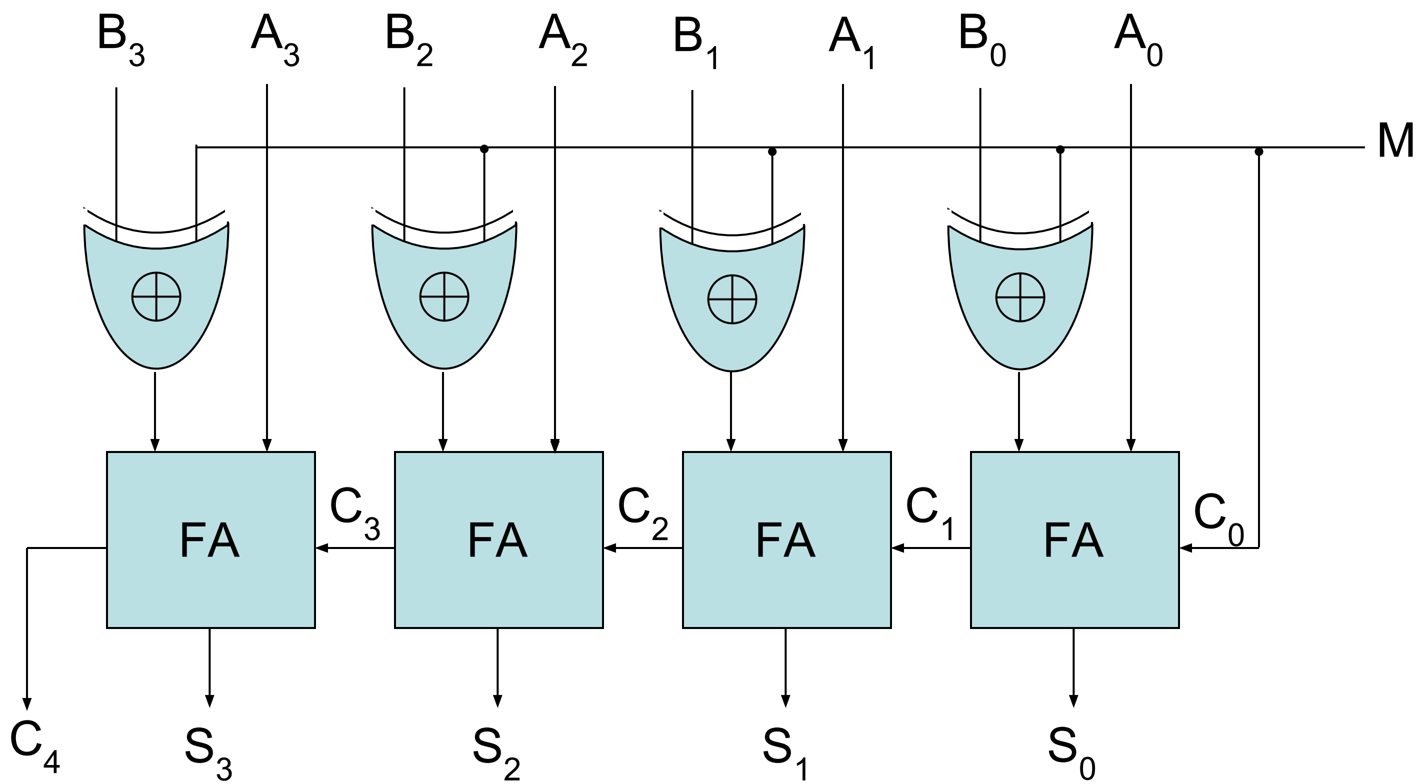


1-4 Arithmetic Microoperations
Binary Adder



4-bit binary adder
(connection of FAs)

1-4 Arithmetic Microoperations
Binary Adder-Subtractor



4-bit adder-subtractor

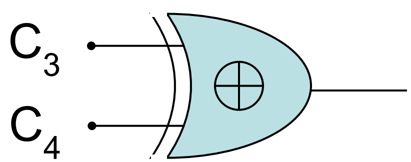
1-4 Arithmetic Microoperations
Binary Adder-Subtractor

- For unsigned numbers, this gives A – B if A≥B or the 2's complement of (B – A) if A < B

(example: 3 – 5 = -2= 1110)

- For signed numbers, the result is A – B provided that there is no overflow. (example : -3 – 5= -8)

$$\begin{array}{r} 1101 \\ 1011\ + \\ \hline 1000 \end{array}$$



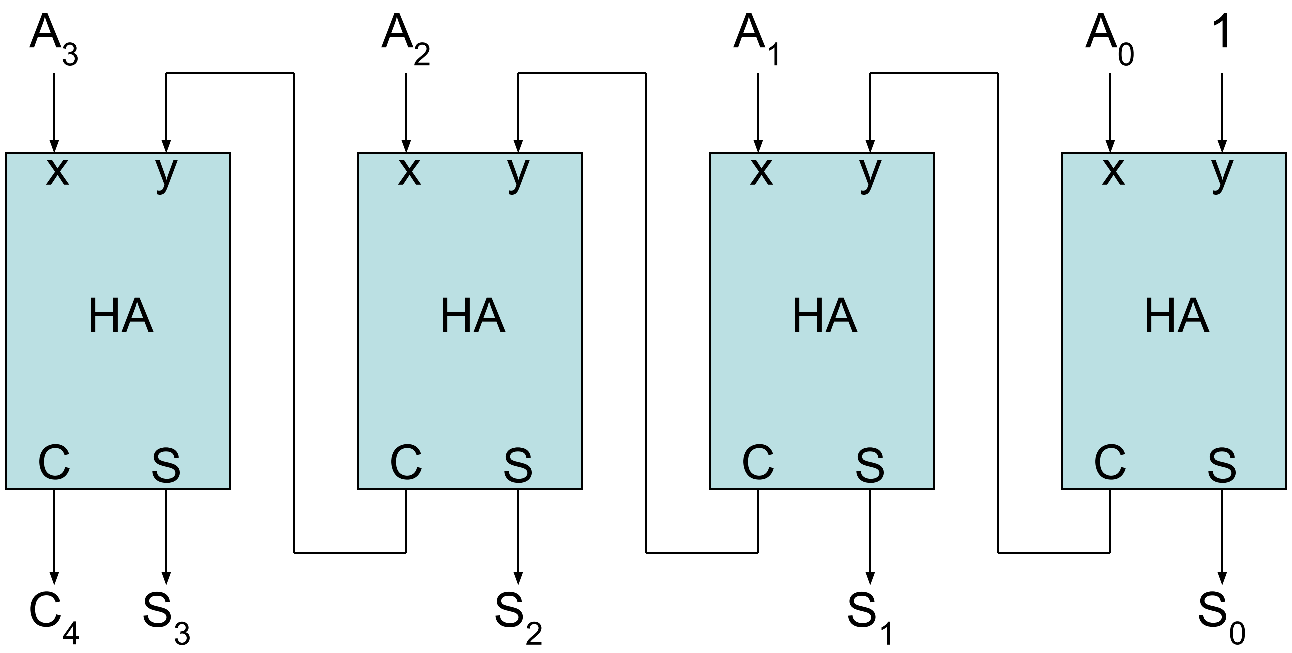
$= V \begin{cases} 1, \text{ if overflow} \\ 0, \text{ if no overflow} \end{cases}$

Overflow detector for signed numbers

1-4 Arithmetic Microoperations Binary Adder-Subtractor cont.

- What is the range of unsigned numbers that can be represented in 4 bits?
- What is the range of signed numbers that can be represented in 4 bits?
- Repeat for n-bit?!

1-4 Arithmetic Microoperations Binary Incrementer



4-bit Binary Incrementer

1-4 Arithmetic Microoperations Binary Incrementer

- Binary Incrementer can also be implemented using a counter
- A binary decrementer can be implemented by adding 1111 to the desired register each time!

- Register transfer microoperations
- Arithmetic microoperations (on numeric data stored in the registers)
- **Logic microoperations (bit manipulations on non-numeric data)**
- **Shift microoperations**

1-4 Arithmetic Microoperations Arithmetic Circuit

- This circuit performs **seven distinct arithmetic operations** and the basic component of it is the **parallel adder**
- The output of the binary adder is calculated from the following arithmetic sum:
  - **$D = A + Y + C_{in}$**

1-4 Arithmetic Microoperations Arithmetic Circuit cont.

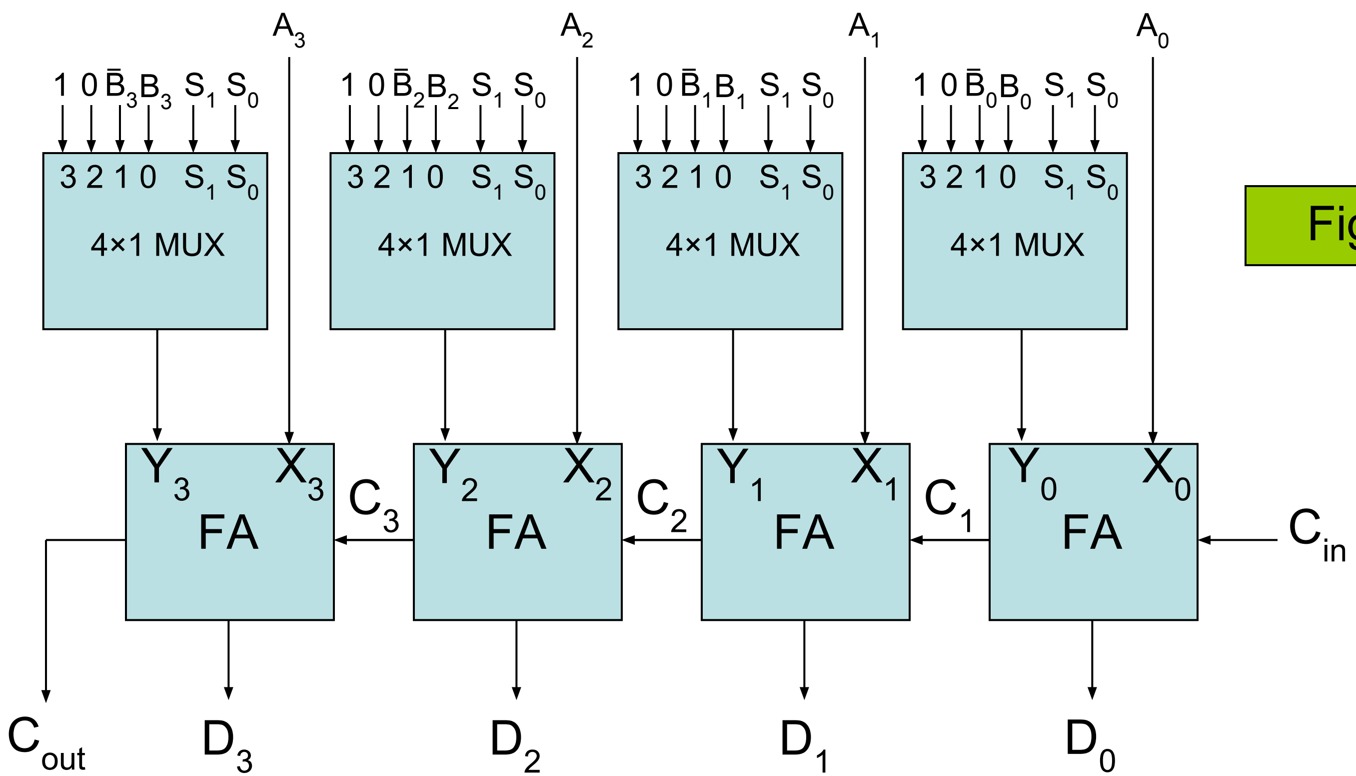


Figure A

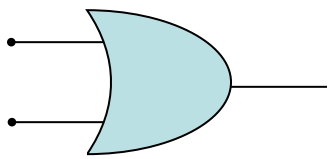
4-bit Arithmetic Circuit

1-4 Arithmetic Microoperations Arithmetic Circuit cont.

Select			Input	Output	
S_1	S_0	C_{in}	Y	$D = A + Y + C_{in}$	Microoperation
0	0	0	B	$D = A + B$	Add
0	0	1	B	$D = A + B + 1$	Add with carry
0	1	0	$\overline{B}$	$D = A + \overline{B}$	Subtract with borrow
0	1	1	$\overline{B}$	$D = A + \overline{B} + 1$	Subtract
1	0	0	0	$D = A$	Transfer A
1	0	1	0	$D = A + 1$	Increment A
1	1	0	1	$D = A - 1$	Decrement A
1	1	1	1	$D = A$	Transfer A

1-5 Logic Microoperations
The four basic microoperations

OR Microoperation

- Symbol: $\vee$, +
- Gate: 
- Example: $100110_2 \vee 101011_2 = 101111_2$

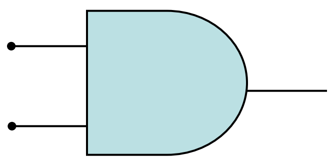
OR　　　　　　　　　　　　　　　　OR

$$P+Q: R1 \leftarrow R2+R3,\ R4 \leftarrow R5 \vee R6$$

ADD

1-5 Logic Microoperations The four basic microoperations
cont.

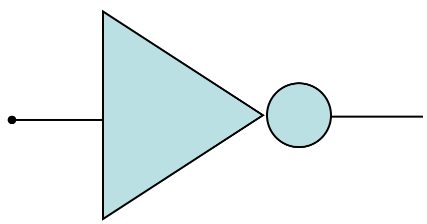
AND Microoperation

- Symbol: $\wedge$
- Gate: 
- Example: $100110_2 \wedge 101011_2 = 100010_2$

1-5 Logic Microoperations The four basic microoperations
cont.

Complement (NOT) Microoperation

- Symbol: $\overline{}$
- Gate:

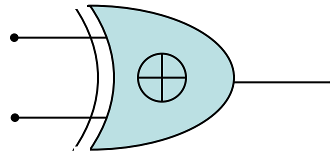


- Example: $\overline{1010110}_2 = 0101001_2$

1-5 Logic Microoperations
The four basic microoperations
cont.

XOR (Exclusive-OR) Microoperation

- Symbol: ⊕
- Gate:



- Example: $100110_2 \oplus 101011_2 = 001101_2$

1-5 Logic Microoperations
Other Logic Microoperations

Selective-set Operation

- Used to force selected bits of a register into logic-1 by using the OR operation

- Example: $0100_2 \vee 1000_2 = 1100_2$

In a processor register

Loaded into a register from memory to perform the selective-set operation

1-5 Logic Microoperations
Other Logic Microoperations cont.

Selective-complement (toggling) Operation

- Used to force selected bits of a register to be complemented by using the XOR operation

- Example: $0001_2 \oplus 1000_2 = 1001_2$

In a processor register

Loaded into a register from memory to perform the selective-complement operation

1-5 Logic Microoperations
Other Logic Microoperations cont.

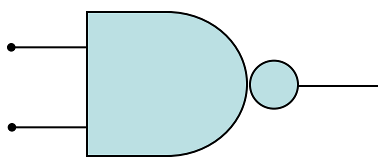
Insert Operation

- Step1: mask the desired bits
- Step2: OR them with the desired value

- Example: suppose R1 = 0110 1010, and we desire to replace the leftmost 4 bits (0110) with 1001 then:
  - Step1: 0110 1010 $\wedge$ 0000 1111
  - Step2: 0000 1010 $\vee$ 1001 0000
- □ R1 = 1001 1010

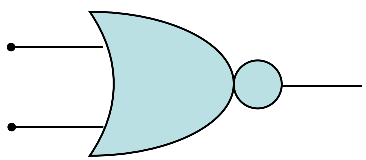
1-5 Logic Microoperations
Other Logic Microoperations
cont.

NAND Microoperation

- Symbols: $\wedge$ and $\overline{}$
- Gate: 
- Example: $\overline{100110_2 \wedge 101011_2} = 011101_2$

1-5 Logic Microoperations Other Logic Microoperations
cont.

NOR Microoperation

- Symbols: $\vee$ and $\overline{}$
- Gate: 
- Example: $\overline{100110_2 \vee 101011_2} = 010000_2$

1-5 Logic Microoperations Other Logic Microoperations
cont.

Set (Preset) Microoperation

- Force all bits into 1's by ORing them with a value in which all its bits are being assigned to logic-1
- Example: $100110_2 \vee 111111_2 = 111111_2$

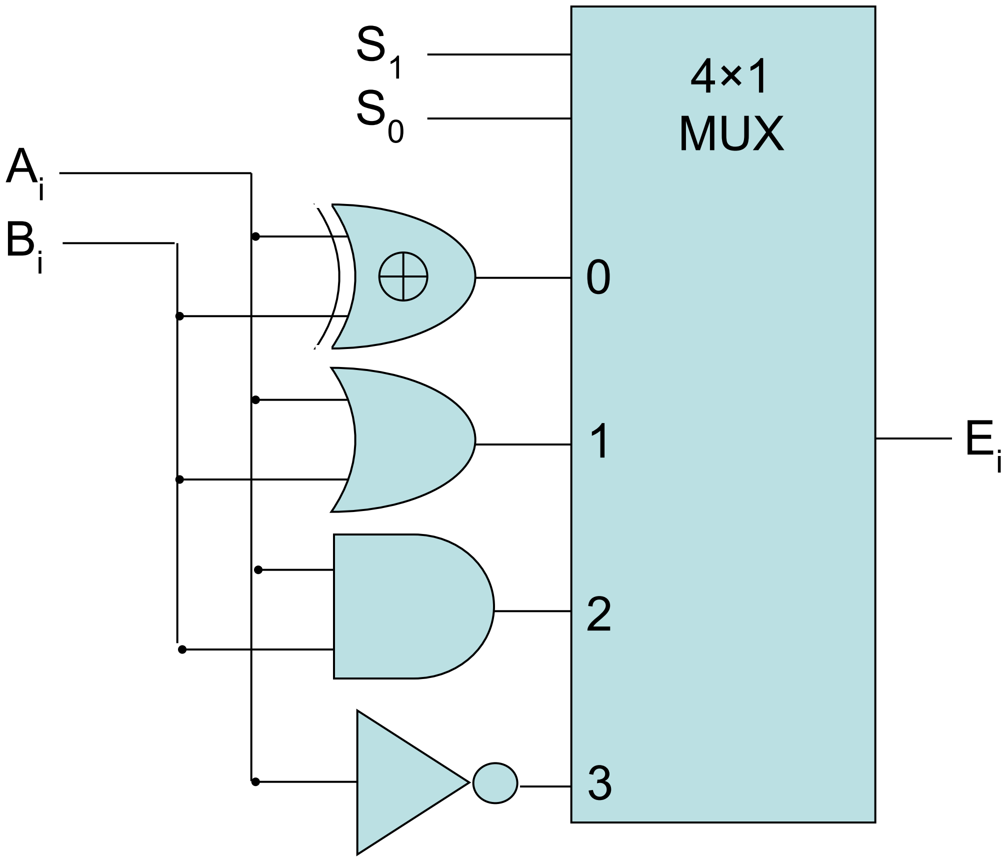
Clear (Reset) Microoperation

- Force all bits into 0's by ANDing them with a value in which all its bits are being assigned to logic-0
- Example: $100110_2 \wedge 000000_2 = 000000_2$

1-5 Logic Microoperations Hardware Implementation

- The hardware implementation of logic microoperations requires that logic gates be inserted for each bit or pair of bits in the registers to perform the required logic function
- Most computers use only four (AND, OR, XOR, and NOT) from which all others can be derived.

1-5 Logic Microoperations Hardware Implementation cont.



S_1	S_0	Output	Operation
0	0	$E = A \oplus B$	XOR
0	1	$E = A \vee B$	OR
1	0	$E = A \wedge B$	AND
1	1	$E = A$	Complement

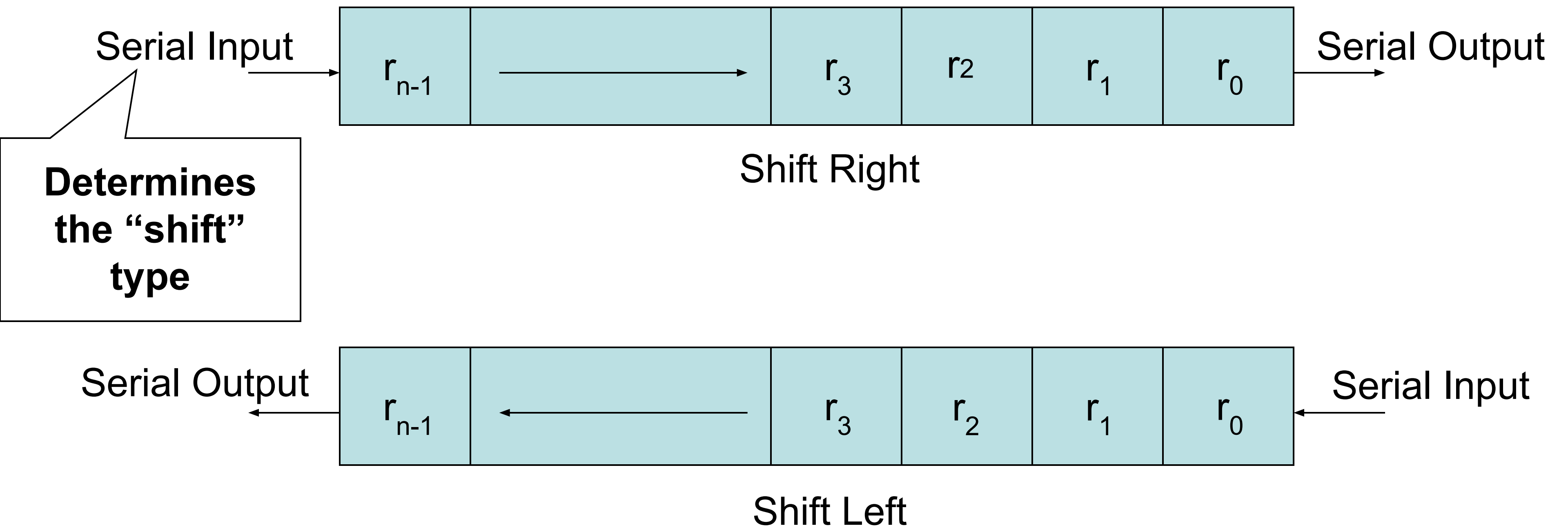
This is for one bit i

Figure B

1-6 Shift Microoperations

- Used for serial transfer of data
- Also used in conjunction with arithmetic, logic, and other data-processing operations
- The contents of the register can be shifted to **the left** or to **the right**
- As being shifted, the first flip-flop receives its binary information from the serial input
- **Three types of shift: Logical, Circular, and Arithmetic**

1-6 Shift Microoperations cont.



**Note that the bit ri is the bit at position (i) of the register

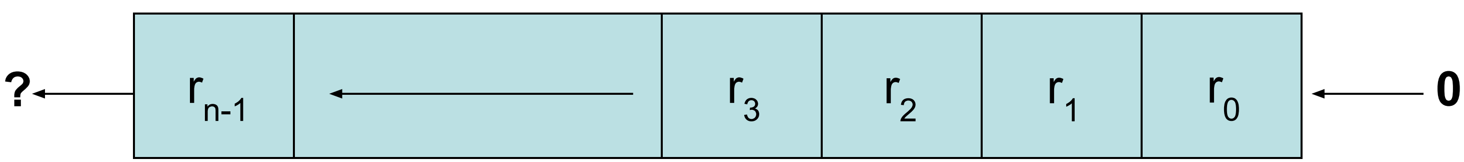
1-6 Shift Microoperations: Logical Shifts

- Transfers 0 through the serial input
- Logical Shift Right: $R1 \leftarrow shr\ R1$

The same

- Logical Shift Left: $R2 \leftarrow shl\ R2$

The same



Logical Shift Left

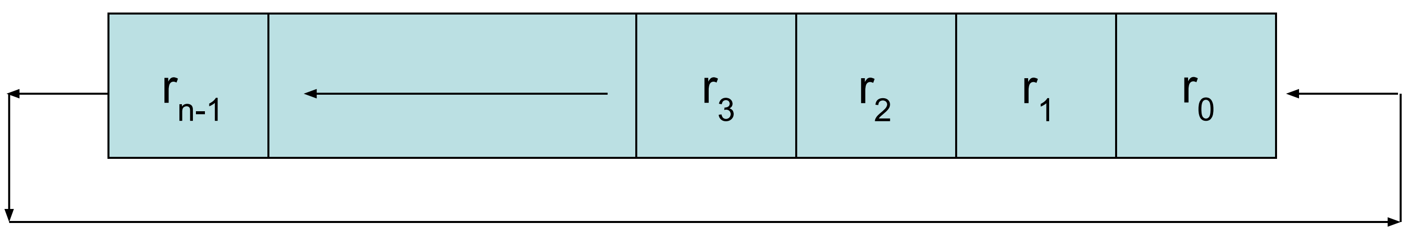
1-6 Shift Microoperations: Circular Shifts (Rotate Operation)

- Circulates the bits of the register around the two ends without loss of information
- Circular Shift Right: R1←cir R1

The same

- Circular Shift Left: R2←cil R2

The same

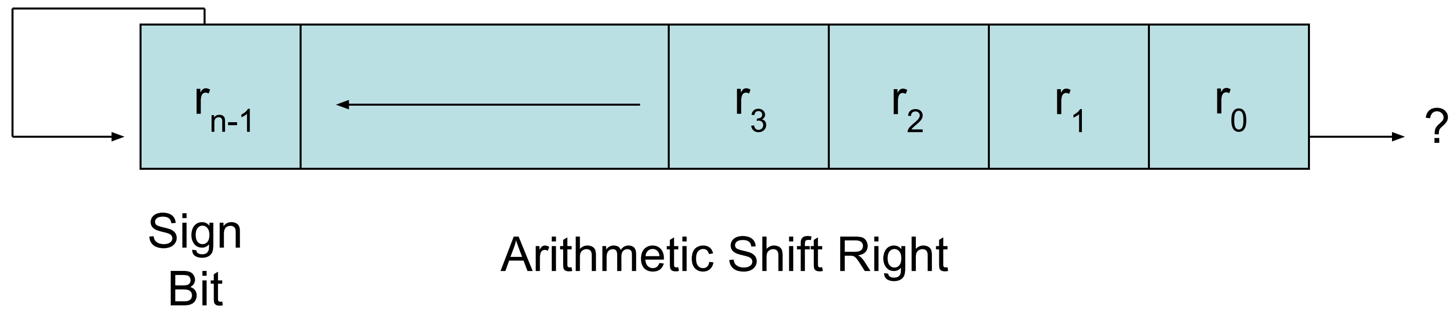


Circular Shift Left

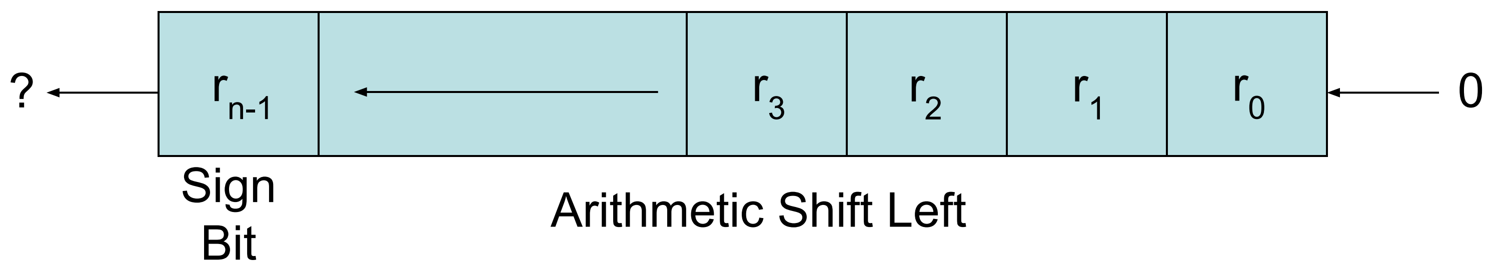
1-6 Shift Microoperations
Arithmetic Shifts

- Shifts a signed binary number to the left or right
- An arithmetic shift-left multiplies a signed binary number by 2:   ashl (00100): 01000
- An arithmetic shift-right divides the number by 2
  ashr (00100) : 00010
- An overflow may occur in arithmetic shift-left, and occurs when the sign bit is changed (sign reversal)

1-6 Shift Microoperations
Arithmetic Shifts cont.



Arithmetic Shift Right

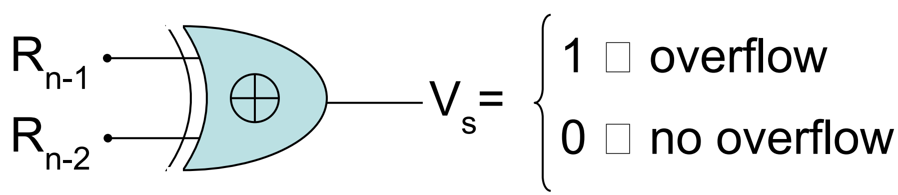


Arithmetic Shift Left

1-6 Shift Microoperations Arithmetic Shifts cont.

- An overflow flip-flop V_s can be used to detect an arithmetic shift-left overflow

$$V_s = R_{n-1} \oplus R_{n-2}$$



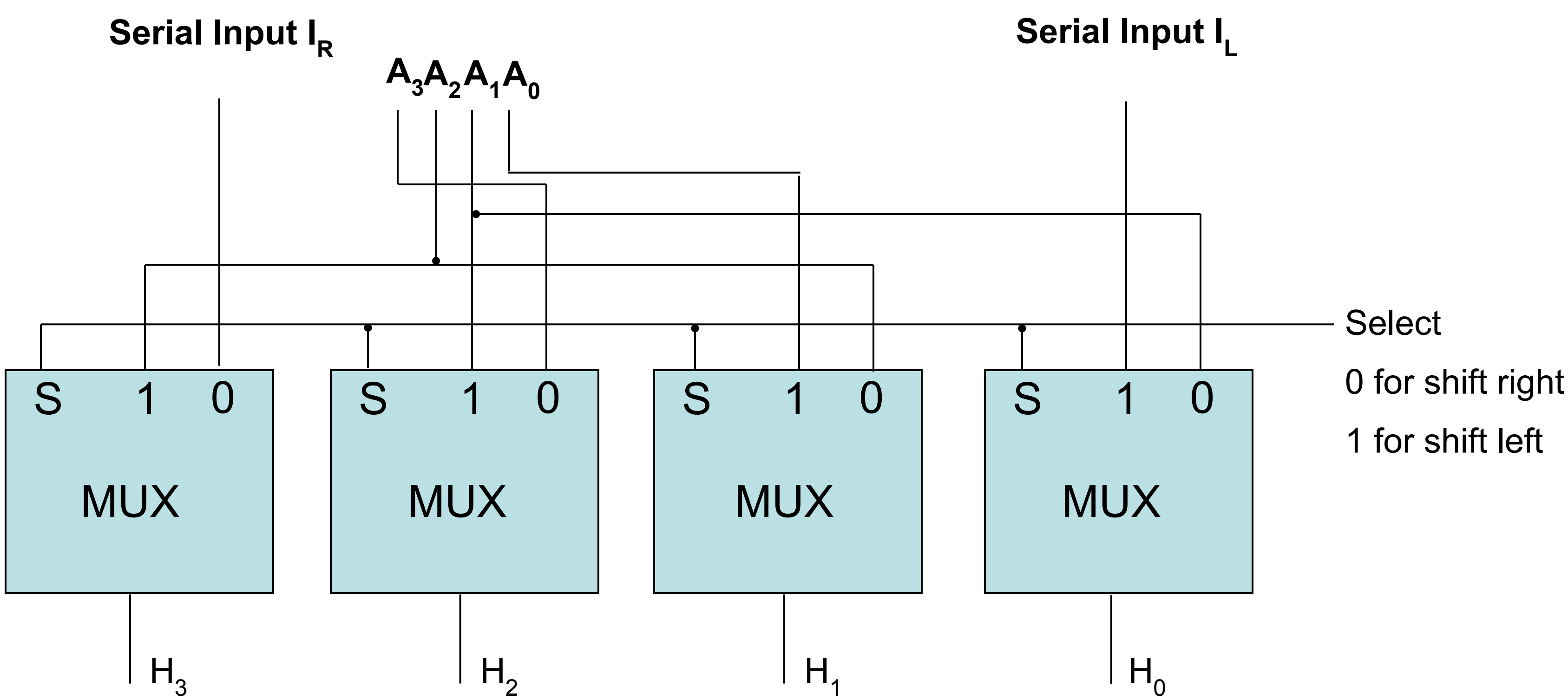
1-6 Shift Microoperations cont.

- Example: Assume R1=11001110, then:
  - Arithmetic shift right once : R1 = 11100111
  - Arithmetic shift right twice : R1 = 11110011
  - Arithmetic shift left once : R1 = 10011100
  - Arithmetic shift left twice : R1 = 00111000
  - Logical shift right once : R1 = 01100111
  - Logical shift left once : R1 = 10011100
  - Circular shift right once : R1 = 01100111
  - Circular shift left once : R1 = 10011101

1-6 Shift Microoperations Hardware Implementation cont.

- A possible choice for a shift unit would be a bidirectional shift register with parallel load (refer to Fig 2-9). Has drawbacks:
  - **Needs two pulses (the clock and the shift signal pulse)**
  - **Not efficient in a processor unit where multiple number of registers share a common bus**
- It is more efficient to implement the shift operation with a combinational circuit
- **PAGE 69**

1-6 Shift Microoperations Hardware Implementation cont.



4-bit Combinational Circuit Shifter

1-7 Arithmetic Logic Shift Unit

- Instead of having individual registers performing the microoperations directly, computer systems employ a number of storage registers connected to a common operational unit called an Arithmetic Logic Unit (**ALU**)

1-7 Arithmetic Logic Shift Unit cont.

One stage of ALU